



Liquid Collective LsETH SY Audit Report

Dec 4, 2025





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Summary

This report has been prepared for Liquid Collective smart contract, to discover issues and vulnerabilities in the source code of their Smart Contract as well as any contract dependencies that were not part of an officially recognized library. A comprehensive examination has been performed, utilizing Static Analysis and Manual Review techniques.

The auditing process pays special attention to the following considerations:

- Testing the smart contracts against both common and uncommon attack vectors.
- Assessing the codebase to ensure compliance with current best practices and industry standards.
- Ensuring contract logic meets the specifications and intentions of the client.
- Cross referencing contract structure and implementation against similar smart contracts produced by industry leaders.
- Thorough line-by-line manual review of the entire codebase by industry experts.



Overview

Project Summary

Project Name	Liquid Collective
Codebase	https://github.com/liquid-collective/pendle-sy-tests
Commit	5f5e5bf3969baad502f39a54f863c0eb8c40eb1b
Language	Solidity

Audit Summary

Delivery Date	Dec 4, 2025
Audit Methodology	Static Analysis, Manual Review
Total Issues	3

[WP-M1] `assetInfo()` should be ETH instead of LsETH

Medium

Issue Description

Currently, both SY's accounting unit (i.e., `yieldToken`) and `asset` are set to `LSETH`, but based on the implementation of `exchangeRate()`, we believe the `asset` should be ETH (the underlying asset of LsETH).

https:

[//docs.pendle.finance/pendle-v2/Developers/Contracts/UnitAndDecimals#conversion-rates](https://docs.pendle.finance/pendle-v2/Developers/Contracts/UnitAndDecimals#conversion-rates)

`SY.exchangeRate()` - returns the exchange rate between SY and the underlying asset.

[https://github.com/liquid-collective/pendle-sy-tests/blob/](https://github.com/liquid-collective/pendle-sy-tests/blob/f902382e66afb93192b27d463c55f6a21f5b9907/src/PendleLsETHETHSY.sol#L11C2-L98C5)

[f902382e66afb93192b27d463c55f6a21f5b9907/src/PendleLsETHETHSY.sol#L11C2-L98C5](https://github.com/liquid-collective/pendle-sy-tests/blob/f902382e66afb93192b27d463c55f6a21f5b9907/src/PendleLsETHETHSY.sol#L11C2-L98C5)

```

11  constructor() SYBaseUpg(LSETH) {}
12
    @@ 13,44 @@
45
46
47  /// @notice Returns the ETH-per-LsETH exchange rate
48  /// @dev This is monotonic (never decreases) as staking rewards accrue
49  /// @return The amount of ETH (1e18 scaled) that 1 LsETH is worth
50  function exchangeRate() public view virtual override returns (uint256) {
51      return ILsETH(LSETH).underlyingBalanceFromShares(1e18);
52  }
53
54
    @@ 55,94 @@
95
96  function assetInfo() external view returns (AssetType assetType, address
assetAddress, uint8 assetDecimals) {
97      return (AssetType.TOKEN, LSETH, IERC20Metadata(LSETH).decimals());
98  }

```



Status

✓ Fixed



[WP-L2] Recommendation: Use Updated `SYBaseUpgV2`

Low

Issue Description

PendleLsETHETHSY is currently using `SYBaseUpg` .

Consider migrating to `SYBaseUpgV2` or `PendleERC20SYUpgV2` .

<https://github.com/pendle-finance/Pendle-SY-Public/blob/main/contracts/core/StandardizedYield/v2/SYBaseUpgV2.sol>

<https://github.com/pendle-finance/Pendle-SY-Public/blob/main/contracts/core/StandardizedYield/implementations/PendleERC20SYUpgV2.sol>

Inheriting `PendleERC20SYUpgV2` will make the code much simpler as most of the code is the same, besides `exchangeRate()` and `assetInfo()` .

Status

✓ Fixed



[WP-N3] Naming conventions: Inconsistency between SY name and yieldToken name

Issue Description

- SY name: "SY Pendle LsETH"
- yieldToken name: "Liquid Staked ETH"

SY symbol "SY-LsETH" matches yieldToken symbol "LsETH".

Using "SY Liquid Staked ETH" as the SY name might better adhere to the SY name conventions.

The current contract name and filename of `PendleLsETHETHSY` is also unconventional, the name should be `PendleLsETHSY` without the last `ETH` (which we assume is the `asset`).

<https://github.com/liquid-collective/pendle-sy-tests/blob/f902382e66afb93192b27d463c55f6a21f5b9907/src/PendleLsETHETHSY.sol#L8-L16>

```
8      address public constant LSETH = 0x8c1BE5b9a0928467c9B1341Da1D7BD5e10b6549; //
      Liquid Collective LsETH on Ethereum mainnet
9
10
11     constructor() SYBaseUpg(LSETH) {}
12
13
14     function initialize() external initializer {
15         __SYBaseUpg_init("SY Pendle LsETH", "SY-LsETH");
16     }
```

Status

✓ Fixed



Appendix

Timeliness of content

The content contained in the report is current as of the date appearing on the report and is subject to change without notice, unless indicated otherwise by WatchPug; however, WatchPug does not guarantee or warrant the accuracy, timeliness, or completeness of any report you access using the internet or other means, and assumes no obligation to update any information following publication.



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